

and

$$\int N e^{3a} G dt = \frac{2}{N^3} e^{3a} \left(\dot{a} + \dot{\beta}_+ \right) \left(2\dot{a} - \dot{\beta}_+ - \dot{\beta}_- \right) \left(2\dot{a} - \dot{\beta}_- + \dot{\beta}_+ \right). \quad (11)$$

2.2. Minisuperspace description

By replacing (10) and (11) in the Action Integral (1) and integrating by parts we end with the following point-like Lagrangian function

$$\begin{aligned} L \left(N, a, \dot{a}, \beta_{\pm}, \dot{\beta}_{\pm} \right) = & \frac{e^{3a}}{N} \left(-3\dot{a}^2 + \frac{3}{4}\dot{\beta}_+^2 + \frac{1}{4}\dot{\beta}_-^2 - 2\Lambda N \right) \\ & + \left(\frac{e^{3a}}{N^3} \dot{\phi} \left(\dot{a} + \dot{\beta}_+ \right) \left(2\dot{a} - \dot{\beta}_+ - \dot{\beta}_- \right) \left(2\dot{a} - \dot{\beta}_- + \dot{\beta}_+ \right) \right). \end{aligned} \quad (12)$$

We remark that when $\dot{\phi} = 0$, i.e. $\phi = 0$, then the limit of General Relativity is recovered.

The field equations follow from the variation of the aforementioned point-like Lagrangian function with respect to the dynamical variables N , a , $\beta_{\pm}$ and ϕ .

Without loss of generality we assume the lapse function $N = 1$. Thus, for the constant lapse function the gravitational field equations read

$$0 = 3H^2 - \frac{3}{4}\dot{\beta}_+^2 - \frac{1}{4}\dot{\beta}_-^2 - 3\dot{\phi} \left(H + \dot{\beta}_+ \right) \left(2H - \dot{\beta}_+ - \dot{\beta}_- \right) \left(2H - \dot{\beta}_- + \dot{\beta}_+ \right) - 2\Lambda, \quad (13)$$

$$\begin{aligned} 0 = & 2\dot{H} + H^2 \left(3 - 4\ddot{\phi} \right) - 2\Lambda + \dot{\beta}_+^3 \dot{\phi} - 8H\dot{H}\dot{\phi} - \dot{\beta}_+ \dot{\phi} \left(\dot{\beta}_-^2 - 2\ddot{\beta}_+ \right) \\ & + \dot{\beta}_+^2 \left(\frac{3}{4} + \ddot{\phi} \right) + \frac{1}{12}\dot{\beta}_- \left(\dot{\beta}_- \left(3 + \ddot{\phi} \right) + 8\dot{\phi}\ddot{\beta}_- \right), \end{aligned} \quad (14)$$

$$\begin{aligned} 0 = & -\frac{3}{2}\ddot{\beta}_+ + 18H^2\dot{\beta}_+\dot{\phi} + 2\dot{\phi} \left(3\dot{\beta}_+ \left(\dot{H} - \ddot{\beta}_- \right) + \dot{\beta}_-\ddot{\beta}_- \right) + \ddot{\phi} \left(\dot{\beta}_-^2 - 3\dot{\beta}_+^2 \right) \\ & + \frac{3}{2}H \left(2\dot{\phi} \left(\dot{\beta}_- + 2\ddot{\beta}_+ \right) + \dot{\beta}_+ \left(4\ddot{\phi} - 3 \right) - 6\dot{\beta}_+^2\dot{\phi} \right), \end{aligned} \quad (15)$$